

GIST 7130 MODULE 03 BUILDING WEBSITES – THE BASICS

Introduction

This module introduces students to

the recommended text editor (Notepad++) to create a web page and place the web page on the GIST 7130 FT web server.

To achieve this the following will be utilized

- Hypertext markup language (HTML) tags for page structure
- Cascading style sheets (CSS) for style, and
- JavaScript for interactions

Objectives

At the end of this lab students should be able to:

Create

- simple web pages using a plain text editor.

Utilize

- HTML tags, CSS and JavaScript to customize a basic web page.

Identify

- basic web page elements.

Access

- GIST 7130 web server.

Host

- an accessible basic web page on the GIST 7130 web server.

Explain

- how web pages are served to a client.

HTML and JavaScript

Why you should know HTML and JavaScript

GIS is being delivered more and more through the internet.

GIS technical staff often install or customize GIS web software.

GIS technical staff setup/maintain underlying web server programs.



HTML and JavaScript pages are generated by GIS web server software

QGIS Server

GeoServer

MapServer

Mapnik

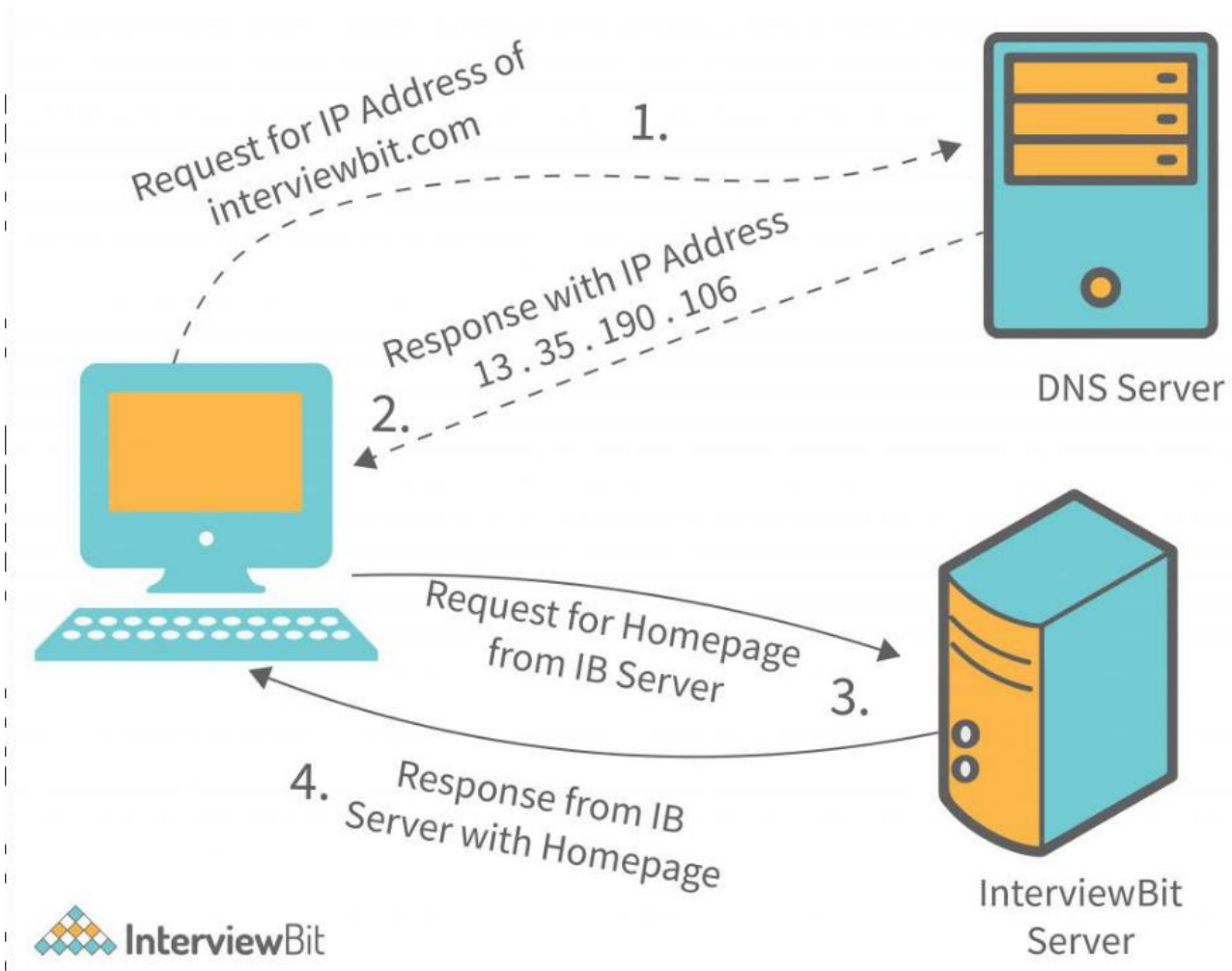
MapGuide

ArcGIS
Server

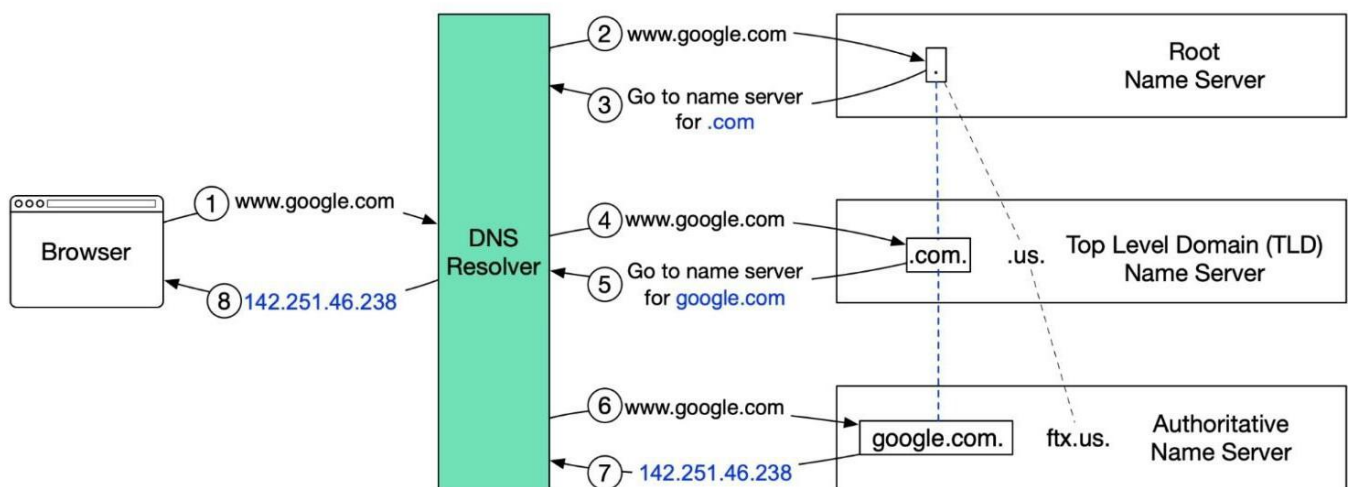


You may have to customize complex web pages for your GIS end-users

GIS technical staff often need to write server-side scripts that generate HTML and JavaScript.



How does DNS resolve IP



EXERCISE 1: REQUEST WEB PAGE**Send Request from Web Client Program**

Open a **web client program** (web browser) and enter the following URL
<https://gisweb.soce.bcit.ca/>

Computer 142.232.230.251 Receives Request with Web Server Program

Web server program listening on port 80 will respond by sending back the default web page (class page) since you did not name a particular web page

Receives Request with Web Client Program

Receives the data from the web server

Attempts to display data in a window (web browser).

Web client programs must be programmed to interpret file format it received.

Web client programs are only **pre-programmed to “understand” certain types of file formats** such as "HTML" files or "PNG" images

You can add on extra programs to web clients, so they understand other type of files such as PDF

Usually, the **web client program just receives text or image data** from the web server and displays it in a window.

Web Server default files

All web client requests

must refer to a specific computer file in a specific disk directory on the web server or the server will send back a "The page cannot be found" error message.

When you go to www.google.ca

you are asking the web server program on the computer at the IP address for www.google.ca to get its default file from the default directory and send the file to you.

All web servers

are configured to have a default directory and a default file name

The default files and directories can be changed, but typically are:

Server Software	Default Directory location
Apache (UNIX)	/var/www/html
Windows	c:\apache\htdocs

The GIS Student home pages are served by a **Windows web server** on a windows server machine.

The default file name has been set to be: **index.html**

So, for example, when you enter the web address:
<http://gisweb.soce.bcit.ca>

you are in fact sent the HTML disk file
[\\student.fs.bcit.ca\GIST_Share\GIS_HTML\index.html](http://student.fs.bcit.ca/GIST_Share/GIS_HTML/index.html)
from the computer gisweb.soce.bcit.ca

Web Pages

are plain **text files**.

Web Servers

deliver web pages to web browsers.

Web pages must either be

pre-written and saved on the web server; OR
created "on-the-fly" by a web server application program (typically written in a programming language like PHP, JavaScript, PERL, and more)

Web pages contain

text and HTML tags which describe how to display the text, and optional programs often written in JavaScript.

JavaScript programs

are used to **make web pages more interactive** and to read and display data.

GIS Web Applications (ArcGIS Online)

mostly **execute on the web server** while the web browser just handles the display with some complex JavaScript code.

Web Browsers

are just software

This means that each web browser **may interpret HTML tags differently**

This is a major complaint of web page designers, for example, the web page may look perfect in Chrome and muddled or unreadable in other browsers.

Lab Module Objective

You **create a web page** in this lab module the "manual way" using a text editor so you can learn the HTML tags and see how the layout affects the result.

You **will have to know HTML tags** when you create or modify custom GIS web clients for your end users.

The text editor

You must use a **basic text editor** when creating and editing your web pages (for this class)

To create a *plain text file*, use a plain text editor.

Plain text editors are **basic editors containing a limited number of functions** and options (unlike advanced editors e.g. Microsoft Word).

Plain Text Editor Options

The **Windows** operating system provides notepad as a plain text editor.

MacOS provides a basic editor called TextEdit.

Advanced Text Editor Options

A very usable free download slightly more advanced text editor is **Notepad++ (windows)**.

Another very popular editor is **Visual Studio Code (also called VS Code)**.

VS Code is an open-source and lightweight text editor available on Windows, Mac, and Linux.

After the basic web page creation

you can use a **website builder** or **web design software** (not this course).

For this lab it is recommended

use **Notepad++** from appsanywhere (or locally),
 save your files to your **BCIT Z:\ drive**, complete the exercise on Z:\ drive then transfer your
 files to your personal student web directory on the GIS server mapped to the **W:\ drive**

Notepad++ or VS Code very useful for writing HTML code

Some useful features are

- **Colour codes** web page elements.
- **Links elements together** so you can see if you have correct opening and closing tags.
- If you double click on a value, **it highlights all instances** of the value within document.
- **You can launch a page** in any of the installed web browsers.

Validation

- **Use Chrome** to display your file in a web browser – this is where pages will be graded from.
- **Use a syntax checker** on your web page.
- **Recommended checker** W3C Mark-up Validation Service <http://validator.w3.org/>
- **You will need to upload the html file** you are checking.

Student web directories

- **Student folders** created based on course enrollment
- **Students have read/write access to their own directory**, and no access to anything else
- **Instructors/admins can read and write** to student directories
- **The web directories can be accessed** from the labs or from off campus (VPN needed) via
 \\student.fs.bcit.ca\GIST_Share\GIS_HTML
- **You must have your web pages in your web directory** so the gisweb web server can retrieve them and deliver them.

EXERCISE 2: YOUR WEB DIRECTORY (OPTIONAL REVIEW)

Review module 01 network locations and module 2 how to map gisweb exercise.



Make sure you can access your web directory.

EXERCISE 3: A BASIC HTML TUTORIAL**Creating Web Pages**

For these exercises the instructions **use Notepad++ from AppsAnywhere** saved on BCIT CITRIX servers

On your BCIT Z:\ drive

Create a folder 7130\mod03

Download all exercise material

- **Save** it in your Z:\7130\mod03 folder (directory)

Start your text editor (VS Code or Notepad++)

Save file as (names must be lowercase) **myfirstpage.html** in your mod03 folder

Complete

'**HTML Beginner Tutorial**' from <http://htmldog.com/guides/html/> Note: change the words of the tutorial – the important part is learning the structure of the document and common tags.

To test

View your pages from **Notepad++** select run -> **launch in Chrome**.

Note: We will look at forms in more detail in module 5.

EXERCISE 4: CASCADING STYLE SHEETS TUTORIAL

Creating CASCADING STYLE SHEETS

When creating web pages the content and the style should be separate.

CSS is used for setting the style – or appearance of a web page and/or site.

How content is displayed e.g. **color, font, backgrounds and so on** should be defined in the CSS file.

With HTML5 and CSS3 **this is the standard**.

All exercise material

is **saved** on your z:\ drive. Then you will upload to server.

Start your text editor (VS Code or Notepad++)

Create new file with Notepad++ and save the file as (names must be lowercase) **style.css** in your **mod03** folder

Complete

'CSS Beginner Tutorial' from <http://html5dog.com/guides/css/>

EXERCISE 5: INTERMEDIATE HTML AND CSS TUTORIAL

From the Intermediate HTML tutorial read/complete Span and Div, Meta Tags and Sectioning.



From the CSS Intermediate tutorial read/complete 'Class and ID Selectors and page layout'.

EXERCISE 6: JAVASCRIPT TUTORIAL

JavaScript

lets you add interaction and dynamic elements to your web page.

You can create popup messages, scrolling messages, animations and complex GIS display programs.

For example: to display data from ArcGIS Server you can use the ArcGIS API for JavaScript, to create a map application for Desktop and Mobile Devices.

With Google Maps you could use the Google Maps JavaScript API, Leaflet Javascript and many other GIS web servers all use JavaScript extensively so you must really know JavaScript to customize any of these products.

JavaScript can be added to your HTML document in different ways:

Inline code – very short sections of JavaScript imbedded within the HTML page.

Javascript in the head section and called from within the HTML page

HTML Javascript event calls

JavaScript saved in an external file with a js extension then reference in the head of the HTML page

The Javascript Date() object is used to work with dates and times.

JavaScript statements end with a semicolon;

Javascript is case sensitive.

For these exercises

Continue to modify **myfirstpage.html**.

Add inline JavaScript code

HTML DOM Document Object

when HTML document is loaded into a web browser, it **becomes a document object**.

The **document object is the root node** of the HTML document.

(https://developer.mozilla.org/en-US/docs/Web/API/HTML_DOM_API)

To write text directly to the HTML document

Use the HTML DOM write() method e.g. `document.write("some text");`

Add the following before the closing `</body>` tag

```
<!--Uses JavaScript to display the date and time -->
```

```
<p>
```

```
  <script>
```

```
    document.write("The date is: "+ Date() );
```

```
  </script>
```

```
</p>
```

Save and display/refresh your html page

You should see **current date and time** from server at **bottom of the page**.

HTML DOM document.write is a basic command

It **prints the text inside the quotes** to your page.

The Date () object

finds the **current system date** and **prints the value** to your screen.

Add a JavaScript function

Use a javascript function to change the color of a line of text.

The **BUTTON** action **ONCLICK** calls the **display** function.

The **display** function changes the text identified by the **ID** of a paragraph element.

The **function** will be in the **HEAD** of the **HTML** file and the **function call** will be in the **main body**.

Add this before closing head tag </head>

```
<script>
  function display()
  {
    var col=document.getElementById("chtxt");
    col.style.color="red";
  }
</script>
```

Add this before closing </body> tag

```
<p id="chtxt">
  Click on the button bellow to change text color.
</p>
<button onclick="display()">
  Change Color
</button>
```

Save and Refresh your page

Click on the **button**

Notice the **text** changes **color**

Add a JavaScript event

In this section you will **add an event to an html tag**.

Prompt the **image to change when mouse moves over the image**.

onMouseOver creates an event when mouse is passed over active text or image.

onMouseOut creates an event when mouse is taken off active text or image.

Images are in the mod03 folder

BCIT_MW.jpg and **BCIT_SSEM_AED2025.jpg** in the **Learning Hub** Or YOU can use your own images.

The images **must be saved in your mod03 folder**.

Add this before the closing </body> tag

```
<p>

</p>
```

Save and Refresh your page

Move the mouse over the image

Notice it changes

Move your mouse off the image

Notice it changes back.

You just completed

three simple JavaScript additions to your web page.

☒ An Inline Script ☒ A Function Call ☒ An Event

EXERCISE 7: INSERT A FAVICON**You can add a favorite icon to the browser tab**

This is the image that many web sites have added to 'brand' their tab. You must first create your icon to display, then save icon to your z:\7130\mod03 folder

Create a favicon at

<http://www.favicon.cc/>

Add the code to your page

Add the following code to the head of your HTML file BEFORE the closing `</head>` tag.

`<!-- Must be relative link -->`

`<link rel="shortcut icon" href="favicon.ico">`

Where href= The name of YOUR favicon.ico **you** created.

Save Refresh your page

You should see your favicon on the browser tab. Example shows my favicon.

Complete

Now that you have created myfirstpage.html on a local drive you will move the page to GISWEB and display from a server using a web browser.

EXERCISE 8: DISPLAY PAGE FROM SERVER**Copy entire mod03 folder to GISWEB server**

Go to YOUR directory – click on your name

COPY your mod03 folder from Z:\7130 and **PASTE** in **YOUR W:\ student web directory on the gisweb server**

Use a web browser and view your web page

Enter <http://gisweb.soce.bcit.ca/students/yourname/mod03/myfirstpage.html>

View source of your page

In Chrome press ctrl-u to view source to see how your file was interpreted and help you find 'errors, or typos' if things are not displaying correctly

Warning

When transferring directories or files **make sure of path**.

It is quite common to put a mod03 directory inside an existing mod03 directory – this has caused problems in the past.

1.
2.

EXERCISE 9: VALIDATE YOUR PAGE

Open a web browser

go to **W3C Markup Validation Service** at <https://validator.w3.org/>

Use either 'validate by file upload

You may/will receive errors and warnings. You must fix these errors and warnings.

You should start with the first error

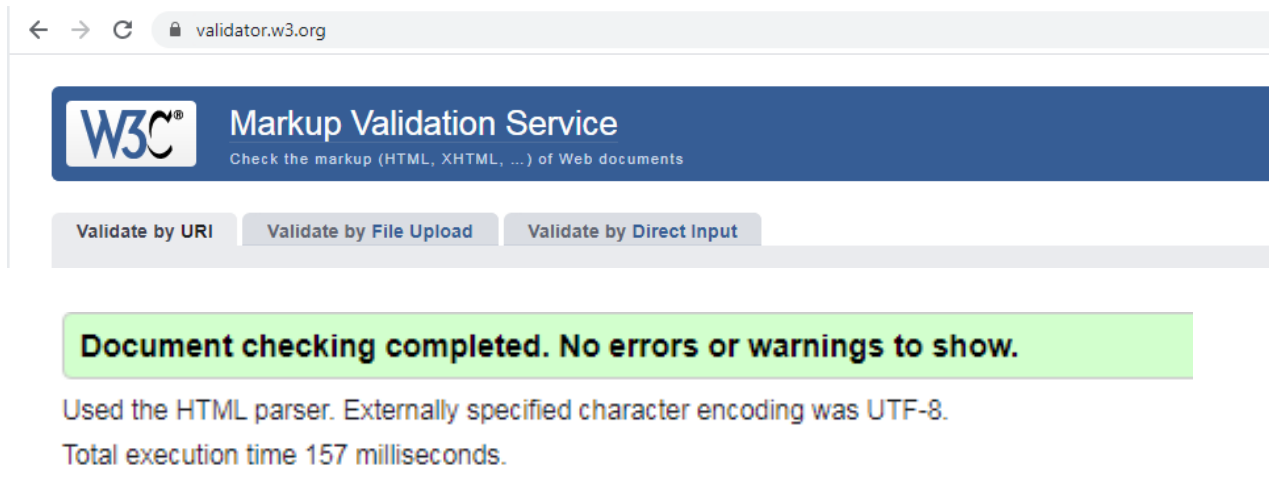
Often the first or second error fixes multiple errors later in the page. Particularly if a quote has been misplaced.

An error you receive if you followed the tutorials

Is related to the language tag. After looking it up using the link change your html tag to: `<html lang="en">`

Save and re validate the page

Continue this process until you receive no errors:



EXERCISE 10: FORMAT CODE

Often when starting with web pages

you will be copying code from various sources. (always make sure to give credit).

When code is copied and pasted

it can appear quite untidy within your document.

You should always

clean up your code for easy reading, debugging and so on.

Use a Web Page Formatter

There are many different options on the web.

HTML Formatter from - <https://webformatter.com/html> can help with this.

Webformatter also will clean up CSS and JavaScript.

See below for example of good formatting.

Example of good formatting: removed extra spaces, consistent indenting, easier to read.

```

1  <!DOCTYPE html>
2
3  <html>
4    <head>
5      <title>My first web page</title>
6      <meta charset="utf-8" />
7      <!-- This is a comment, by the way -->
8      <link rel="stylesheet" href="style.css" />
9      <script>
10         function display() {
11             var col = document.getElementById("chtxt");
12             col.style.color = "red";
13         }
14     </script>
15     <!-- Must be relative link -->
16     <link rel="shortcut icon" href="favicon.ico" />
17 </head>

```

For this course

you can choose the indenting spacing but be consistent in your document(s)

Use Webformater

to clean up your HTML page and CSS file. [Or a formatter of your choice]

After you clean your code

make sure the updated file is placed in your web directory.

You now know how to:

1. create an html text document
2. enter HTML tags
3. create styles for your HTML document
4. transfer your files to your web directory on the gisweb server
5. display your web page.
6. validate and fix errors on your web page
7. create consistent formatted code

ASSIGNMENT SPECIFICATIONS

For this one-week lab assignment

answer a question in **Learning Hub** Quiz and
save web pages in your **BCIT GIS WEB** account

All pages *must*

be **named as specified**
name the page **mypage.html**
be in the **correct location(s)**

All files *must*

be in your web **mod03** directory within the correct subdirectories(s) and
have the **correct name(s)**

Page *must*

Be accessible from your **web mod03** directory.
References external style sheet named **mystyle.css**
References a custom **myfavicon**.

Assignment Objective

Create a **basic professional web page** about you and/or your interest(s) **that demonstrates you can apply the basic concepts** as described in tutorials

Authenticity

do not use tutorial colors or text
be custom to YOU and or your interests
use a plain text editor

THINGS THAT ARE BEING ASSESSED

HTML structure: correct tags, tags open and closed properly

CSS in external style sheet

Page validates without errors or warnings.

Favicon displays.

Inclusion of a table with data

JavaScript Image(s) minimum of 2

Heading present

Inclusion of a list

Overall document(s) format is clean and professional

Text and color selection is accessible

There should be no extra code and indenting should be consistent.

This is a basic – ‘nice’ page – to get started.

SUMMARY

After completing the first 3 modules students should be able to create an HTML web page with simple HTML, JavaScript and CSS. This module oriented each student with:

Saving files on custom student web page directory

How web page files are served by the `gisweb.soce.bcit.ca`

Creating simple web pages using a plain text editor and linking it to a css stylesheet

Creating a css stylesheet using a plain text editor

Using HTML tags to define the layout hyperlinks to images and other web pages.

Using JavaScript to make the web page more interactive.